

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 4 Maths — CA-1 (Hard)

Class : 4

Subject : Maths (Math-Mela)

Max Marks : 70

Duration : 2 Hr 30 Min

Syllabus: Chapters 1–3 — Shapes Around Us, Hide and Seek, Pattern Around Us

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

SECTION A — Multiple Choice Questions (1 Mark each)

- A solid shape has 5 faces, 9 edges, and 6 corners. Using Faces + Corners - Edges = 2, verify if this could be a valid solid, and identify it. (1 Mark)
A. $5 + 6 - 9 = 2$, valid; likely a triangular prism
B. $5 + 6 - 9 = 0$, invalid
C. $5 + 9 - 6 = 8$, invalid
D. The formula cannot be used here
- If you see only the top view and front view of an object but not the side view, what limitation might you face in fully understanding its shape? (1 Mark)
A. No limitation, two views are always enough
B. You might not know the exact depth or shape variations only visible from the side
C. Top and front views are identical, so no information is lost
D. It becomes impossible to know anything about the object
- A sequence of numbers goes 2, 4, 6, 8, What is the 10th number in this pattern? (1 Mark)
A. 18
B. 20
C. 22
D. 16
- Which of these combinations of coins/notes to make ₹8377;145 uses the FEWEST number of pieces (available: 100, 50, 20, 10, 5)? (1 Mark)
A. $100 + 20 + 20 + 5$
B. $50 + 50 + 20 + 20 + 5$
C. $100 + 50 - 5$ (not valid)
D. $20 \times 7 + 5$
- A pyramid has a square base. How many triangular faces will it have in addition to its base? (1 Mark)
A. 2
B. 3
C. 4
D. 5
- If a grid position is described as "2 rows down and 3 columns right" from the top-left corner, which of these best describes this position? (1 Mark)
A. Row 2, Column 3
B. Row 3, Column 2
C. Row 1, Column 1
D. It cannot be determined
- Which statement is TRUE about the sum of two odd numbers? (1 Mark)
A. The sum is always odd
B. The sum is always even
C. The sum can be either odd or even
D. Odd numbers cannot be added together
- A cube-shaped box is cut in half straight through the middle, parallel to one face. What shape will the cut cross-section be? (1 Mark)
A. Circle
B. Square
C. Triangle
D. Pentagon
- If Priya has ₹350 and wants to buy an item worth ₹275, how much change should she receive, and using the fewest notes/coins (100, 50, 20, 10, 5), how could this change be given? (1 Mark)
A. ₹75 as $50+20+5$
B. ₹85 as $50+20+10+5$
C. ₹75 as $20 \times 3 + 10 + 5$
D. ₹65 as $50+10+5$

SECTION B — Short Answer Questions (2 Marks each)

- Explain why knowing only the top view of a building is not enough to fully construct it, using reasoning about missing information. (0 Marks)
- A pattern of numbers goes 3, 7, 11, 15, Identify the rule of the pattern and find the next two numbers. (0 Marks)
- Explain why a triangular pyramid (with a triangular base) has fewer faces than a square pyramid (with a square base), using their properties. (0 Marks)
- A group of 47 students is being divided for two different games - one needing pairs, and one needing groups of 5. Explain which grouping will have students left over and why. (0 Marks)
- Explain how understanding both grid positions and directions (like up, down, left, right) together helps in describing movement, such as in a simple maze. (0 Marks)
- A cube net (unfolded cube) is shown with 6 squares arranged in a cross shape. Explain how you would verify that this net can actually fold into a cube. (0 Marks)
- Explain why the statement "all shapes with 4 sides are squares" is incorrect, using at least two counter-examples. (0 Marks)

8. A vending machine only accepts exact change for a 65¢ item. Explain, using available denominations of 50¢, 20¢, 10¢, 5¢, how many different combinations could make exactly 65¢. (0 Marks)
9. Explain why understanding both 2D shape properties and 3D shape properties together is important when studying real architecture, using an example. (0 Marks)
10. A number pattern skips every third number: 1, 2, 4, 5, 7, 8, Explain the rule and predict the next three numbers. (0 Marks)
11. Explain the concept of symmetry in relation to shapes like a square, and why a scalene triangle (with all different side lengths) generally lacks this property. (0 Marks)
12. Explain how a "bird's eye view" (top view) used in maps differs conceptually from a "cross-sectional view" (a view showing what's inside if something were cut open). (0 Marks)
13. Explain why the concept of "fewest coins/notes" when making change is considered an efficient strategy, and describe one situation where this might not be practical. (0 Marks)
14. A 3D shape has exactly one flat face and one curved surface, meeting at a single point. Identify the shape and explain your reasoning. (0 Marks)

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. Analyse how the properties of 3D shapes (faces, edges, corners) follow a predictable mathematical relationship ($\text{Faces} + \text{Corners} - \text{Edges} = 2$), and discuss why this rule is useful for verifying whether a described shape is possible. (0 Marks)
2. Discuss how combining knowledge of views (top, front, side) with grid-based positioning creates a more complete spatial understanding than either skill alone. (0 Marks)
3. Critically evaluate why understanding even and odd number patterns, though seemingly simple, forms an important foundation for more advanced mathematical concepts learned in later years. (0 Marks)
4. Discuss the practical and mathematical value of learning to make payments using the fewest possible notes/coins, connecting this to broader real-world financial literacy. (0 Marks)
5. Analyse how the three chapters (Shapes Around Us, Hide and Seek, Pattern Around Us) together build a foundation for spatial reasoning and logical thinking, which are important for subjects beyond mathematics. (0 Marks)
6. Discuss whether learning about 3D shapes primarily through classroom activities (like building models) is more effective than learning purely through textbook diagrams, and explain your reasoning. (0 Marks)

SECTION D — Long Answer Questions (5 Marks each)

1. Write a detailed essay analysing how the three chapters studied (Shapes Around Us, Hide and Seek, Pattern Around Us) connect to real-world careers or fields, such as architecture, design, or data analysis, explaining the specific skills from each chapter that would be useful. (0 Marks)
2. Design a small "maths trail" activity for younger students, using at least one concept each from "Shapes Around Us," "Hide and Seek," and "Pattern Around Us." Describe the activity in detail and explain how it teaches each concept. (0 Marks)
3. Reflect on which of the three chapters (Shapes Around Us, Hide and Seek, Pattern Around Us) you found most interesting or useful, and write a detailed explanation of why, including specific examples from the chapter and how you might use this knowledge in your own life. (0 Marks)